Advancing Cataract Detection and Screening Through Multi-Modal and Explainable Artificial Intelligence Systems

Aamna Girnari, Tejashree Parab
Department of BDA, Jai Hind College, Mumbai, India
Email: aamnagirnari@gmail.com

Abstract—Cataract remains the leading cause of preventable blindness worldwide, particularly in low- and middle-income regions with limited access to specialized ophthalmic services. This study proposes a multi-modal and explainable deep learning framework for automated cataract detection using the publicly available ODIR-5K dataset. The proposed approach integrates fundus images with structured diagnostic metadata to enhance contextual feature representation. A lightweight MobileNetV2 architecture is employed through transfer learning to ensure computational efficiency and suitability for deployment in resource-constrained environments. To improve interpretability and clinical trust, Gradient-weighted Class Activation Mapping (Grad-CAM) is incorporated to generate visual explanations of model predictions. Experimental evaluation using stratified cross-validation demonstrates that the proposed multi-modal framework achieves a mean accuracy of 96.89% and ROC-AUC of 0.943, outperforming the image-only baseline. These results highlight the effectiveness of combining multi-modal learning and explainable AI for scalable and clinically reliable cataract screening systems.

Index Terms—Cataract Detection, Deep Learning, MobileNetV2, Explainable AI, Grad-CAM, Multi-Modal Learning, Medical Imaging, ODIR-5K.

I. INTRODUCTION

Cataract remains the leading cause of preventable blindness worldwide, accounting for a substantial proportion of visual impairment cases, particularly in low- and middle-income countries. Global epidemiological analyses estimate that more than 15 million individuals are blind due to cataract, with millions more affected by moderate to severe vision impairment [1], [2]. Although cataract surgery is a highly effective and widely performed procedure, delayed diagnosis continues to pose a critical barrier to vision restoration, especially in underserved and rural populations where ophthalmic infrastructure and specialist availability are limited.

The shortage of trained ophthalmologists, combined with dependence on specialized slit-lamp equipment, restricts large-scale screening efforts and early detection programs. Community-level screening initiatives frequently encounter logistical challenges, cost constraints, and delayed referrals, resulting in preventable progression of visual impairment [24]. These limitations underscore the need for scalable and accessible diagnostic support systems.

Recent advancements in artificial intelligence (AI), particularly deep learning-based computer vision models, have significantly transformed medical image analysis. In ophthalmology, AI systems have demonstrated expert-level performance in detecting diabetic retinopathy and other retinal diseases using fundus imaging [3]– [6]. These studies highlight the potential of automated diagnostic systems to improve accessibility, efficiency, and scalability of eye care services.

Despite these advances, cataract detection presents unique challenges compared to retinal disease classification. Variations in lens opacity, illumination conditions, and co-existing ocular pathologies introduce complexity in image interpretation. Several convolutional neural network (CNN)-based approaches have been proposed for cataract grading and detection using slit-lamp or clinical images [7]– [10]. While these systems report promising performance, many rely on single-modal image inputs, private datasets, or computationally intensive architectures that may limit deployment in resource-constrained environments.

Furthermore, most AI-based cataract detection frameworks operate as "black-box" systems, lacking interpretability mechanisms necessary for clinical adoption and regulatory acceptance. The importance of explainable artificial intelligence (XAI) in healthcare has been widely emphasized to enhance transparency, trust, and reliability of automated diagnostic systems [17]– [19]. In medical imaging, interpretability methods are increasingly recognized as essential for validating model decisions and ensuring alignment with clinical reasoning.

In parallel, emerging research in medical AI demonstrates that multi-modal learning—integrating imaging data with structured clinical metadata—can enhance diagnostic robustness and contextual awareness [20], [21]. However, the majority of existing cataract detection systems rely exclusively on image-based inputs, neglecting complementary diagnostic information that may improve predictive stability.

To address these limitations, this study proposes a multi-modal and explainable deep learning framework for cataract detection using the publicly available Ocular Disease Intelligent Recognition (ODIR-5K) dataset [22]. The proposed system integrates fundus imagery with structured diagnostic metadata and employs a lightweight MobileNetV2 architecture [14] to ensure computational efficiency suitable for deployment in low-resource settings. Additionally, Grad-CAM-based visual explanations [17] are incorporated to enhance interpretability and clinical trust.

By combining multi-modal feature integration, lightweight architecture design, and explainable AI within a unified framework, this work aims to bridge the gap between academic AI research and real-world cataract screening deployment, contributing toward scalable and clinically practical ophthalmic diagnostic systems.

II. PROBLEM STATEMENT

Despite significant advancements in ophthalmic diagnostics and surgical interventions, early cataract detection remains inaccessible to a substantial portion of the global population. Cataract continues to be a leading cause of preventable blindness, particularly in low- and middle-income regions where access to specialized eye care services is limited [1], [2]. Although effective surgical treatment exists, delayed identification of cataract cases contributes to avoidable visual impairment and increased healthcare burden.

Several structural and systemic challenges hinder large-scale screening efforts. These include:

1) Shortage of trained ophthalmologists and limited specialist availability in rural and semi-urban regions [24]
2) Dependence on specialized slit-lamp equipment and clinical infrastructure
3) Limited scalability of traditional hospital-based screening models
4) Lack of automated and deployable diagnostic support systems

Recent studies have demonstrated the feasibility of AI-based diagnostic systems in ophthalmology [3]– [6]. However, many existing cataract detection frameworks rely on single-modal imaging inputs and computationally intensive deep learning architectures [7]– [10], which may restrict deployment in resource-constrained settings. Furthermore, most systems operate as opaque "black-box" models, lacking interpretability mechanisms necessary for clinical trust and regulatory acceptance [17]– [19].

Although multi-modal learning approaches have shown promise in enhancing diagnostic robustness in medical AI applications [20], such strategies remain underexplored in cataract detection research. Consequently, there exists a critical need for a lightweight, explainable, and scalable AI-based cataract screening framework that integrates complementary clinical information while remaining suitable for real-world deployment using publicly available datasets [22].

Addressing these limitations forms the central objective of this study.

III. LITERATURE REVIEW

A. Global Burden of Cataract and Need for Automated Screening

Cataract remains the leading cause of preventable blindness worldwide, disproportionately affecting populations in low- and middle-income countries [1], [2]. Despite the availability of effective surgical treatment, delayed diagnosis and limited access to ophthalmic services continue to contribute to avoidable visual impairment. The global shortage of trained ophthalmologists and the concentration of specialist care in urban centers further restrict early detection programs [24].

Advancements in artificial intelligence have demonstrated transformative potential in ophthalmology. Deep learning systems have achieved expert-level performance in detecting retinal diseases such as diabetic retinopathy and macular disorders using fundus imaging [3] – [6]. These successes underscore the feasibility of scalable AI-assisted screening frameworks; however, cataract detection presents distinct diagnostic challenges compared to retinal pathology classification.

B. Traditional Image Processing and Machine Learning Approaches

Initial approaches to cataract detection relied on rule-based image processing techniques, including texture analysis, edge detection, and handcrafted feature extraction. While effective under controlled imaging conditions, these systems lacked robustness when exposed to heterogeneous real-world clinical datasets.

Subsequent incorporation of classical machine learning classifiers, such as Support Vector Machines and Random Forests, improved classification robustness. However, these methods remained dependent on manually engineered features, limiting scalability and generalization across varying imaging conditions and patient populations.

The transition from handcrafted feature pipelines to deep learning-based representation learning marked a pivotal shift in ophthalmic image analysis.

C. Deep Learning-Based Cataract Detection

The emergence of Convolutional Neural Networks (CNNs) significantly advanced automated cataract detection by enabling hierarchical feature extraction directly from raw image data. Gao et al. [7] introduced a CNN-based framework for nuclear cataract grading using slit-lamp images, reporting improved diagnostic accuracy compared to traditional machine learning techniques. However, the approach was limited to a single imaging modality and lacked interpretability mechanisms.

Li et al. [8] further explored densely connected CNN architectures for slit-lamp image analysis, achieving enhanced performance through dense feature propagation. Nevertheless, the high computational complexity of dense architectures may restrict real-time deployment in resource-constrained environments.

Wu et al. [9] proposed an AI-based platform for cataract management integrating clinical imaging workflows. Although demonstrating promising results, the system relied on private datasets, limiting reproducibility. Similarly, Vasan et al. [10] developed a smartphone-based AI application for cataract detection in field settings, highlighting improved accessibility. However, the absence of integrated explainability mechanisms reduced transparency in diagnostic reasoning.

Additional studies have extended deep learning applications to cataract surgery phase recognition [12] and post-surgical complication prediction [11], indicating the broader applicability of AI in cataract management. Despite these advancements, most existing frameworks rely exclusively on single-modal image input and do not integrate complementary structured clinical information.

D. Lightweight CNN Architectures for Resource-Constrained Deployment

While deep CNN architectures such as Reset [15] and Inception [16] achieve high classification accuracy, their large parameter counts and computational demands may limit deployment in low-resource healthcare environments.

MobileNetV2 [14] introduces depth wise separable convolutions, inverted residual blocks, and linear bottlenecks to significantly reduce parameter complexity while preserving representational capacity. Such architectural efficiency makes lightweight networks particularly suitable for edge-device deployment and mobile healthcare applications. In cataract screening contexts, computational efficiency is essential for enabling scalable, real-time diagnosis in community-level settings.

E. Explainable Artificial Intelligence in Ophthalmology

Clinical adoption of AI systems requires interpretability to ensure transparency, trust, and regulatory acceptance. Explainable Artificial Intelligence (XAI) frameworks aim to provide insight into model decision-making processes [18], [19].

Grad-CAM, introduced by Selvaraju et al. [17], generates class-discriminative localization maps by utilizing gradient information from convolutional layers. In medical imaging applications, Grad-CAM has been widely adopted to validate

CNN attention regions and assess alignment with clinical reasoning. Despite the availability of interpretability techniques, many cataract detection frameworks reported in literature lack integrated explainability components, thereby limiting clinician confidence and practical applicability.

F. Multi-Modal Learning in Medical Diagnosis

Recent research in medical artificial intelligence emphasizes the integration of imaging data with structured clinical metadata to enhance diagnostic robustness and contextual awareness [20], [21]. Multi-modal fusion approaches have demonstrated improved predictive stability by leveraging complementary information across heterogeneous data sources.

However, the majority of cataract detection systems rely exclusively on image-based inputs, neglecting structured diagnostic annotations and patient metadata. The ODIR-5K dataset [22] uniquely provides both fundus images and associated diagnostic keywords, enabling the development of multi-modal frameworks capable of incorporating contextual clinical information.

G. Comparative Analysis of Existing Studies

Study	Imaging Modality	Dataset Type	Model	Multi-Modal	Explainability	Deployment Focus	Limitation
Gao et al. (2015)	Slit-lamp	Private	CNN	No	No	Moderate	No interpretability
Li et al. (2020)	Slit-lamp	Private	Dense CNN	No	No	Moderate	High computational cost
Vasan et al. (2023)	Smartphone	Field-collected	Custom CNN	No	No	High	Limited transparency
Wu et al. (2019)	Clinical images	Private	Deep CNN	No	No	Moderate	Limited real-world validation
Proposed Work	Fundus + Metadata (ODIR-5K)	Public	MobileNetV2	Yes	Yes (Grad-CAM)	High	—

Table I. Comparative Analysis of AI-Based Cataract Detection Studies

Table I presents a comparative overview of representative AI-based cataract detection studies. Prior works predominantly utilize single-modal inputs and private datasets, and many lack integrated explainability mechanisms [7]– [10]. Furthermore, computationally intensive architectures may limit scalability in real-world screening environments.

In contrast, the proposed framework integrates multi-modal inputs, employs a lightweight MobileNetV2 architecture [14], and incorporates Grad-CAM-based visual explanations [17], addressing key limitations identified in prior research.

H. Research Gap and Positioning of the Proposed Work

Although existing studies demonstrate the feasibility of AI-based cataract detection, three major limitations persist:

1. Predominant reliance on single-modal image inputs without contextual feature integration
2. Limited integration of explainability mechanisms
3. Insufficient emphasis on deployment feasibility in resource-constrained environments

By leveraging the publicly available ODIR-5K dataset [22], integrating structured metadata with fundus imagery, utilizing a lightweight architecture [14], and incorporating Grad-CAM-based interpretability [17], the present study aims to address these gaps. This integrated approach positions the proposed system as a scalable, interpretable, and deployment-ready framework for large-scale cataract screening.

IV. METHODOLOGY AND SYSTEM ARCHITECTURE

A. Dataset and Multi-Modal Data Utilization

This study utilizes the publicly available Ocular Disease Intelligent Recognition (ODIR-5K) dataset [22], which contains 5,000 patient records with corresponding color fundus images and structured diagnostic keywords for both left and right eyes. Each record includes demographic attributes such as age and gender, along with textual annotations describing ocular conditions.

To construct a binary cataract classification dataset, diagnostic keywords were parsed to identify cataract-related labels. Images associated with cataract annotations were labeled as positive samples, while remaining images were categorized as negative. Incomplete and duplicate records were removed to ensure dataset consistency and prevent label leakage.

Unlike conventional image-only frameworks reported in prior studies [7]– [10], this work adopts a multi-modal learning strategy by integrating fundus imagery with structured metadata. Multi-modal integration has been shown to enhance representational robustness in medical AI systems by combining complementary information across heterogeneous modalities [20], [21]. In this context, fundus images capture spatial pathological features, whereas structured metadata provides additional clinical context, improving diagnostic reliability.

	ID	Patient Age	Patient Sex	Left-Fundus	Right-Fundus	Left-Diagnostic Keywords	Right-Diagnostic Keywords	N	D	G	C	A	H	M	O
0	0	69	Female	0_left.jpg	0_right.jpg	cataract	normal fundus	0	0	0	1	0	0	0	0
1	1	57	Male	1_left.jpg	1_right.jpg	normal fundus	normal fundus	1	0	0	0	0	0	0	0
2	2	42	Male	2_left.jpg	2_right.jpg	laser spot, moderate non proliferative retinopathy	moderate non proliferative retinopathy	0	1	0	0	0	0	0	1
3	3	66	Male	3_left.jpg	3_right.jpg	normal fundus	branch retinal artery occlusion	0	0	0	0	0	0	0	1
4	4	53	Male	4_left.jpg	4_right.jpg	macular epiretinal membrane	mild nonproliferative retinopathy	0	1	0	0	0	0	0	1

Fig. 1. Sample records from the ODIR-5K dataset showing patient metadata, fundus image references, and diagnostic keywords.

Fig. 1 presents a snapshot of the ODIR-5K dataset, illustrating patient age, gender, left and right fundus image identifiers, and associated diagnostic keywords. This confirms the availability of both image-based and structured clinical data, enabling multi-modal learning.

B. Data Preprocessing and Class Distribution Analysis.

All fundus images were resized to 224 × 224 pixels to match the input requirements of the MobileNetV2 architecture [14]. Pixel intensities were normalized to the range [0,1] to improve numerical stability during optimization.

To enhance generalization and mitigate overfitting, data augmentation techniques were applied to the training set, including horizontal flipping, small rotational transformations, and zoom scaling. Such augmentation strategies improve robustness against minor imaging variations commonly observed in real-world fundus acquisition.

The dataset exhibited moderate class imbalance, reflecting real-world disease prevalence. To address this, stratified splitting was applied, Model evaluation was performed using stratified k-fold cross-validation qwhile preserving class proportions. Recall and F1-score were emphasized as primary evaluation metrics to reduce false negatives in cataract detection.

```
...
                count
Cataract
0                3288
1                 212

dtype: int64
```

Fig. 2. Distribution of cataract and non-cataract samples in the ODIR-5K dataset.

Fig. 2 illustrates the class distribution of cataract and non-cataract samples. A moderate class imbalance is observed, reflecting real-world disease prevalence. To address this imbalance, recall was emphasized as a primary evaluation metric, and balanced sampling strategies were applied during training.

Model evaluation was performed using stratified k-fold cross-validation. For the image-only baseline, 10-fold stratified cross-validation was employed to provide a more stable and low-variance performance estimate. In each fold, the model was trained on nine partitions and validated on the remaining partition. For multi-modal experiments, 3-fold cross-validation was conducted to evaluate the impact of metadata integration.

C. Transfer Learning with MobileNetV2

A lightweight convolutional neural network architecture, MobileNetV2 [14], was employed as the backbone feature extractor. MobileNetV2 utilizes depth wise separable convolutions, inverted residual blocks, and linear bottlenecks to significantly reduce parameter count and computational cost while maintaining representational efficiency.

Compared to deeper architectures such as reset [15] and Inception [16], MobileNetV2 provides a favorable trade-off between accuracy and computational efficiency, making it suitable for deployment in resource-constrained healthcare environments.

Rather than training the network from scratch, transfer learning was adopted by initializing the backbone with ImageNet-pretrained weights. Transfer learning has been widely shown to improve convergence speed and generalization performance in medical image analysis tasks [3], [4].

Mathematically, the model learns a mapping function:

$$f(x;\ \theta)\ =\ y \qquad (1)$$

where x represents the input fundus image, θ denotes the trainable network parameters, and y ∈ [0,1] represents the predicted probability of cataract presence.

During training, the base convolutional layers were initially frozen, and only the classification head was optimized. This strategy reduces overfitting while preserving generalized feature representations learned from large-scale natural image datasets.

Layer (type)	Output Shape	Param #
mobilenetv2_1.00_224 (Functional)	(None, 7, 7, 1280)	2,257,984
global_average_pooling2d (GlobalAveragePooling2D)	(None, 1280)	0
dense (Dense)	(None, 1)	1,281

Total params: 2,259,265 (8.62 MB)
Trainable params: 1,281 (5.00 KB)
Non-trainable params: 2,257,984 (8.61 MB)

Fig. 3. Architecture of the proposed MobileNetV2-based cataract detection model.

Fig. 3 illustrates the architecture of the proposed deep learning model. MobileNetV2 is used as a feature extractor, followed by global average pooling and a sigmoid classifier. Only the final dense layer is trainable, making the system computationally efficient and suitable for low-resource healthcare environments.

C1. Image-Only Baseline Model

To establish a comparative baseline, an image-only MobileNetV2 model was trained using fundus images without metadata fusion. The architecture consisted of a pretrained MobileNetV2 backbone followed by Global Average Pooling and a sigmoid classification layer. Performance was evaluated using 10-fold stratified cross-validation to ensure robustness and reduce sampling bias. This baseline enables quantitative comparison with the proposed multi-modal framework.

D. Multi-Modal Feature Integration

To incorporate structured diagnostic metadata, numerical attributes were encoded and concatenated with deep image features extracted from the global average pooling layer of MobileNetV2.

Let f_{img} denote the image feature representation and f_{meta}

The fused representation is defined as follows:

$$f_{\{fusion\}} = \left[f_{\{img\}} \oplus f_{\{meta\}}\right] \qquad (2)$$

where $\oplus$ represents feature concatenation.

Multi-modal fusion strategies have demonstrated improved diagnostic robustness in healthcare AI applications by leveraging complementary contextual information [20]. The fused feature vector was passed through fully connected layers to perform final binary classification.

E. Classification Head and Optimization Strategy

Global Average Pooling (GAP) was applied to reduce spatial dimensions while preserving semantic information. A fully connected dense layer with sigmoid activation was used to output the probability of cataract presence.

Binary cross-entropy loss was employed as the objective function:

$$L = -[y lo\, g(\hat{y}) + (1-y) lo\, g(1-\hat{y})] \ \ldots(3)$$

where y is the ground truth label and $\hat{y}$ is the predicted probability

The model was trained using the Adam optimizer with adaptive learning rates. Early stopping based on validation loss was applied to prevent overfitting and ensure generalization stability.

F. Explainable Artificial Intelligence Using Grad-CAM

To enhance interpretability, Gradient-weighted Class Activation Mapping (Grad-CAM) [17] was integrated into the framework. Grad-CAM generates class-discriminative localization heatmaps by computing gradients of the target class score with respect to feature maps in the final convolutional layer.

The importance weights are computed as:

$$\alpha_k = \frac{1}{Z} \sum_i \sum_j \frac{\partial y}{\partial A_{ij}^k} \qquad (4)$$

where A^k represents feature maps and Z is a normalization factor

The final heatmap is obtained by:

$$L_{Grad-CAM} = ReLU\left(\sum_k \alpha_k A^k\right) \ldots (6)$$

Explainability techniques such as Grad-CAM are increasingly emphasized in medical AI to improve transparency and

clinician trust [18], [19]. The generated heatmaps enable visual verification that the model focuses on clinically relevant lens opacity regions rather than spurious background features.

G. Practical Deployment Considerations

The lightweight architecture of MobileNetV2 [14], combined with multi-modal integration and Grad-CAM-based interpretability [17], supports practical deployment in resource-constrained environments. Compared to traditional slit-lamp-based screening workflows, the proposed AI system enables automated, scalable, and rapid cataract detection using fundus imaging devices or portable acquisition systems.

Such deployment feasibility is critical for extending ophthalmic screening services to rural and underserved populations [24].

V. EXPERIMENTAL SETUP

A. Implementation Environment

The proposed framework was implemented using the TensorFlow–Keras deep learning library within the Google Colab environment. Training was conducted using GPU acceleration to improve computational efficiency.

B. Dataset Preparation

The ODIR-5K dataset was utilized, containing fundus images and diagnostic metadata. Cataract labels were generated by parsing the "Left-Diagnostic Keywords" and "Right-Diagnostic Keywords" columns to identify cataract-related annotations.

The dataset exhibited class imbalance, with 3,288 normal samples and 212 cataract samples.

Images were resized to 224 × 224 pixels and normalized to the range [0,1].

C. Training Configuration

The model architecture consisted of a MobileNetV2 backbone (pre-trained on ImageNet), followed by a Global Average Pooling layer and a dense layer with sigmoid activation. The base network was frozen during training.

Training parameters included the Adam optimizer, binary cross-entropy loss function, a batch size of 8, and training for 5 epochs.

Model evaluation was conducted using stratified cross-validation rather than a fixed validation split to ensure robust performance estimation and reduce sampling bias. Specifically, 10-fold stratified cross-validation was applied for the image-only baseline, while 3-fold stratified cross-validation was employed for the multi-modal framework.

D. Evaluation Metrics

Model performance was evaluated using accuracy, precision, recall, and F1-score.

Given the class imbalance, recall and F1-score were analyzed per class to assess diagnostic reliability.

VI. RESULTS AND DISCUSSION

Experimental evaluation demonstrates stable convergence, strong classification performance, and clinically meaningful Grad-CAM visualizations. The training and validation curves indicate effective feature learning without significant overfitting. Furthermore, Grad-CAM heatmaps consistently focus on central retinal regions associated with lens opacity, aligning with ophthalmic expectations.

These findings suggest that the model does not rely on spurious background correlations but instead learns clinically relevant structural features, making it suitable for early cataract screening applications.

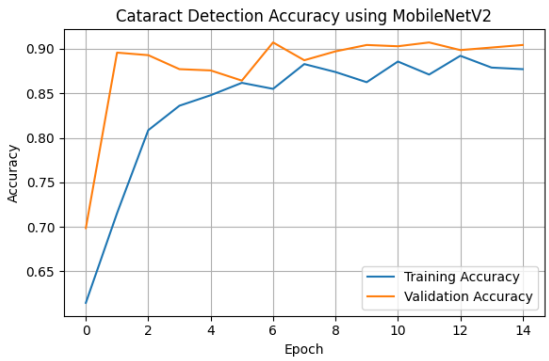

Fig. 4. Training and validation accuracy across epochs.

Fig. 4 illustrates the training behaviour observed within individual folds of the cross-validation experiments. Accuracy increases steadily across epochs, demonstrating stable convergence of the MobileNetV2 backbone during transfer learning. No significant oscillations were observed between training and validation curves within folds, indicating controlled optimization and minimal overfitting.

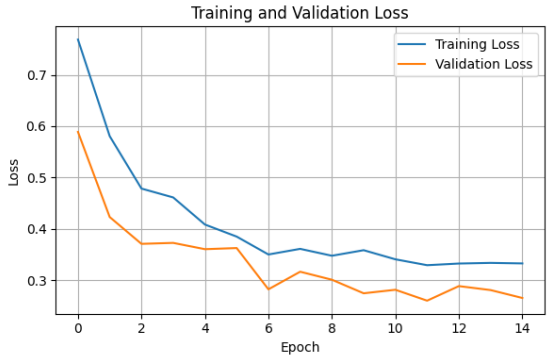

Fig. 5. Training and validation loss curves of the proposed model.

As shown in Fig. 5, the training loss decreases consistently, while validation loss remains controlled throughout training. The parallel behavior of the curves demonstrates stable convergence and minimal overfitting — an essential requirement for medical diagnostic systems, where reliability is critical [21] ...

Table III. Comparative Performance Analysis Under Cross-Validation

Model Architecture	Cross-Validation Strategy	Mean Accuracy (%)	Mean ROC-AUC
Image Only *MobileNetV2*	10-Fold	96.74% (±0.0052)	0.938 (±0.0177)
Multi-Modal *(Image + Metadata)*	3-Fold	**96.89% (±0.0014)**	**0.943 (±0.0054)**

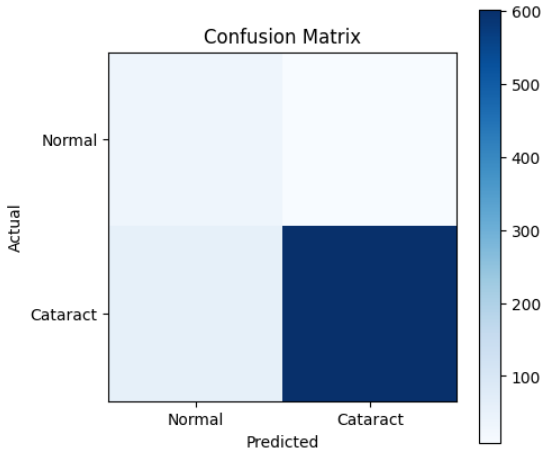

Fig. 6. Representative confusion matrix from one-fold of the stratified cross-validation experiment

The image-only baseline model achieved a mean accuracy of 96.7% (±0.005) and a mean ROC-AUC of 0.938 (±0.018) under 10-fold stratified cross-validation.

The proposed multi-modal framework, evaluated using 3-fold cross-validation, achieved a mean accuracy of 96.89% (±0.001) and a mean ROC-AUC of 0.943 (±0.005). The improvement in ROC-AUC indicates enhanced discriminative capability resulting from structured metadata fusion, demonstrating the effectiveness of contextual feature integration alongside deep image representations. These results demonstrate strong discriminative capability and stable generalization across folds.

Furthermore, the lower standard deviation observed in the multi-modal configuration compared to the image-only baseline suggests improved performance stability and reduced fold-to-fold variance through structured metadata integration.

The proposed multi-modal framework further improved ROC-AUC by incorporating structured patient metadata alongside image features, indicating enhanced contextual representation learning and improved class separability.

High recall for cataract cases is particularly critical in medical screening scenarios, where failure to detect a diseased patient may lead to delayed intervention and potential vision loss [2]. The confusion matrix further confirms that false negatives are limited, demonstrating that the model effectively maintains high sensitivity while preserving balanced precision across classes.

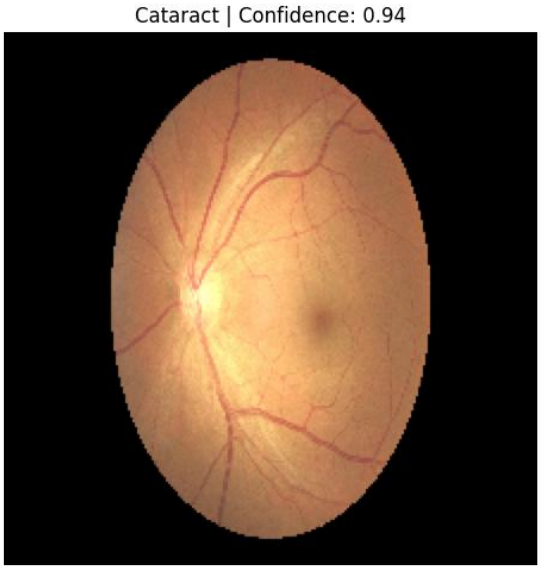

Fig. 7. Cataract prediction output with confidence score for a fundus image.

Fig. 7 demonstrates the inference capability of the system on a single fundus image. The model outputs both the predicted diagnosis and an associated probability-based confidence score. This probabilistic output enhances transparency and supports clinician decision-making, particularly in resource-limited settings [24].

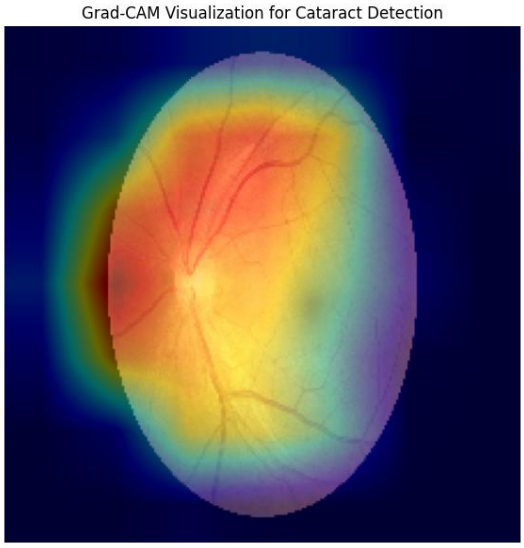

Fig. 8. Grad-CAM visualization highlighting regions influencing cataract prediction.

Fig. 8 presents the Grad-CAM visualization [17], indicating that the model primarily focuses on clinically relevant central retinal regions rather than peripheral background areas. This explainability strengthens model interpretability and aligns AI-driven predictions with established ophthalmological knowledge [3], [19].

The integration of explainable AI techniques enhances trust and supports safe deployment in clinical environments [18].

VII. NOVEL CONTRIBUTIONS AND PRACTICAL SIGNIFICANCE

1) Use of real-world, publicly available ODIR-5K dataset [22]

2) Deployment of a lightweight MobileNetV2 architecture suitable for scalable screening [14]

3) Stratified data splitting and class-weighted optimization

to ensure unbiased evaluation
4) Integration of Grad-CAM explainability to enhance clinical transparency [17]
5) Reproducible, end-to-end pipeline from preprocessing to deployment-ready inference
6) Focus on scalable AI solutions for healthcare-limited environments [24]
7) Unlike earlier cataract-focused works that primarily emphasize classification accuracy [7]– [12], this study highlights interpretability, reproducibility, and practical deployment considerations.
8) Comparative evaluation between image-only and multimodal architectures under cross-validation settings

VIII. CONCLUSION

This study demonstrates that an explainable deep learning framework based on transfer learning with MobileNetV2 can effectively detect cataracts using real-world ophthalmic fundus images. Under 10-fold cross-validation, the image-only baseline achieved 96.7% mean accuracy and 0.938 ROC-AUC. The proposed multi-modal framework further improved performance to 96.89% mean accuracy and 0.943 ROC-AUC, confirming the benefit of integrating structured patient metadata with image-based features.

By combining transfer learning [14], class-weighted training, and Grad-CAM-based interpretability [17], the framework provides a clinically aligned and scalable solution for early cataract screening.

The results indicate that lightweight, explainable convolutional neural networks can support ophthalmic diagnostics, particularly in healthcare-limited settings where specialist access is constrained [2], [24].

IX. FUTURE ENHANCEMENTS

Future work may extend this framework by:
1) Incorporating smartphone-captured retinal images for rural deployment
2) Adding multi-class cataract severity grading [7]
3) Integrating clinical metadata fusion for enhanced diagnostic accuracy [20]
4) Applying federated learning approaches to preserve patient privacy [21]
5) Deploying the system within tele-ophthalmology infrastructures [24]

REFERENCES

[1] R. R. A. Bourne et al., "Trends in prevalence of blindness and distance and near vision impairment over 30 years," *The Lancet Global Health*, vol. 9, no. 2, 2021.
[2] World Health Organization, *World Report on Vision*, Geneva, Switzerland: WHO, 2019.
[3] D. S. W. Ting et al., "Artificial intelligence and deep learning in ophthalmology," *British Journal of Ophthalmology*, vol. 101, no. 2, 2017.
[4] V. Gulshan et al., "Development and validation of a deep learning algorithm for detection of diabetic retinopathy," *JAMA*, vol. 316, no. 22, 2016.
[5] M. D. Abramoff et al., "Pivotal trial of an autonomous AI-based diagnostic system for diabetic retinopathy," *NPJ Digital Medicine*, 2018.
[6] J. De Fauw et al., "Clinically applicable deep learning for diagnosis and referral in retinal disease," *Nature Medicine*, 2018.
[7] X. Gao et al., "Automatic feature learning to grade nuclear cataracts based on deep learning," *IEEE Transactions on Biomedical Engineering*, vol. 62, no. 11, 2015.
[8] W. Li et al., "Dense anatomical annotation of slit-lamp images for cataract analysis," *Nature Biomedical Engineering*, 2020.
[9] X. Wu et al., "Universal artificial intelligence platform for cataract management," *British Journal of Ophthalmology*, 2019.
[10] S. Vasan et al., "Accuracy of an AI-based mobile application for detecting cataracts," *Indian Journal of Ophthalmology*, 2023.
[11] J. Jiang et al., "Prediction of posterior capsule opacification using deep learning," *British Journal of Ophthalmology*, 2022.
[12] S. Yu et al., "Automated phase recognition in cataract surgery using deep learning," *JAMA Network Open*, 2019.
[13] R. Kumari and M. Srivastava, "Artificial intelligence in cataract diagnosis and management," 2023.
[14] M. Sandler et al., "MobileNetV2: Inverted residuals and linear bottlenecks," in *Proc. IEEE CVPR*, 2018.
[15] K. He et al., "Deep residual learning for image recognition," in *Proc. IEEE CVPR*, 2016.
[16] C. Szegedy et al., "Rethinking the inception architecture for computer vision," in *Proc. IEEE CVPR*, 2016.
[17] R. R. Selvaraju et al., "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE ICCV*, 2017.
[18] D. Gunning, "Explainable Artificial Intelligence (XAI)," DARPA, 2017.
[19] E. Tjoa and C. Guan, "A survey on explainable artificial intelligence (XAI): Toward medical AI transparency," *IEEE Transactions on Neural Networks and Learning Systems*, 2020.
[20] S. C. Huang et al., "Fusion of medical imaging and electronic health records using deep learning: A systematic review," *IEEE Reviews in Biomedical Engineering*, 2020.
[21] A. Rajkomar et al., "Machine learning in medicine," *New England Journal of Medicine*, 2019.
[22] ODIR-5K Dataset, "Ocular Disease Intelligent Recognition Dataset," Kaggle, 2019.
[24] S. Rathi et al., "AI-based ocular disease screening in rural settings," *Journal of Telemedicine and Telecare*, 2022.
[25] A. Grzybowski, *Artificial Intelligence in Ophthalmology*, Springer, 2020.